**Check your grasp!**

$C_4$ plants requires 30 ATPs and 12 NADPH + $H^+$ to synthesize one glucose, but $C_3$ plants requires only 18 ATPs and 12 NADPH + $H^+$ to synthesize one glucose molecule. If then, how can you say $C_4$ plants are more advantageous?

Solution: $C_4$ plants are more advantageous than $C_3$ plants because most of the energy lost during photo respiration in $C_3$ plants.

## 13.13 Crassulacean Acid Metabolism or CAM cycle

It is one of the carbon pathways identified in succulent plants growing in semi-arid or xerophytic condition. This was first observed in crassulaceae family plants like *Bryophyllum*, *Sedum*, *Kalanchoe* and is the reason behind the name of this cycle. It is also noticed in plants from other families Examples: *Agave*, *Opuntia*, Pineapple and Orchids. The stomata are closed during day and are open during night (Scotoactive). This reverse stomatal rhythm helps to conserve water loss through transpiration and will stop the fixation of $CO_2$ during the day time. At night time CAM plants fix $CO_2$ with the help of Phospho Enol Pyruvic acid (PEP) and produce oxalo acetic acid (OAA). Subsequently OAA is converted into malic acid like $C_4$ cycle and gets accumulated in vacuole increasing the acidity. During the day time stomata are closed and malic acid is decarboxylated into pyruvic acid resulting in the decrease of acidity. $CO_2$ thus formed enters into Calvin Cycle and produces carbohydrates (Figure13.19).

**Significance of CAM Cycle**

1. It is advantageous for succulent plants to obtain $CO_2$ from malic acid when stomata are closed.
2. During day time stomata are closed and $CO_2$ is not taken but continue their photosynthesis.
3. Stomata are closed during the day time and help the plants to avoid transpiration and water loss.

## 13.14 Photorespiration or $C_2$ Cycle or Photosynthetic Carbon Oxidation (PCO) Cycle

Respiration is a continuous process for all living organisms including plants. **Decker** (1959) observed that rate of respiration is more in light than in dark. Photorespiration is the excess respiration taking place in photosynthetic cells due to absence of $CO_2$

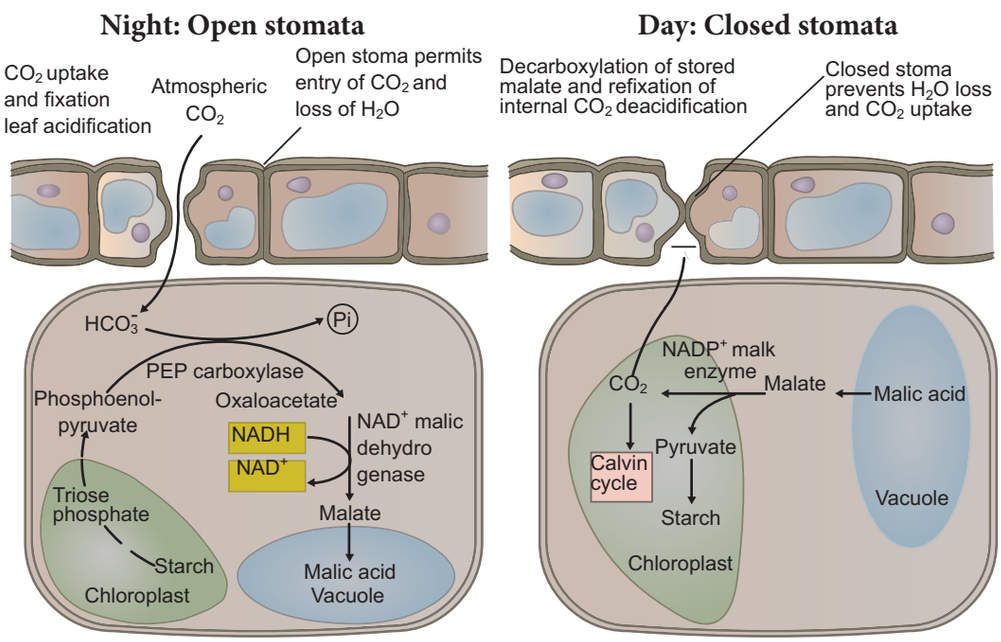



**Figure 13.19:** CAM cycle



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